J-Space and the Chain of Thought: Where Reasoning State Lives

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Abstract

We ask where a language model keeps intermediate results during multi-step reasoning: in J-space, the internal concept workspace read out by the Jacobian lens, or in the written chain of thought. Against the hierarchy hypothesis we wrote down before running, models write out every intermediate step even on trivial problems, fail almost immediately when writing is forbidden, and are unaffected by a J-space ablation calibrated to leave normal text generation intact. Editing and patching experiments show the written chain of thought is where serial computation happens: the model reads its own written values back through the internal state at each written token, the answer follows whatever value we write into that state, and the only state that stays internal is parallel or cheaply recomputable. We report results on Qwen3-4B over GSM8K, MATH-500, and AIME 2024, with synthetic-task experiments isolating the mechanism.

1. Research question and hypothesis

When a language model solves a multi-step problem, where do the intermediate results live: in J-space, the internal concept workspace that the Jacobian lens reads out of the model's activations, or in the chain of thought the model writes? We predicted, before running anything, that models use the internal workspace first and start writing intermediate results down only when problems get hard enough to overflow it. That predicts two things: easy problems should show little writing, and deleting the internal workspace (ablating J-space) should hurt answers produced without chain of thought more than answers produced with it.

Neither prediction held. Models write out every intermediate step even on trivial problems, they fail almost immediately when writing is forbidden, and a J-space ablation gentle enough to leave normal text generation intact changes nothing. What we found instead, through editing and patching experiments, is that the written chain of thought is where the serial computation actually happens: the model reads its own written values back through the internal state at each written token, and the only state that stays internal is either parallel (many values needed at once, none depending on the others) or cheap to recompute from the problem statement. The report shows this on the assignment model and datasets first, then uses synthetic tasks to explain the mechanism behind the benchmark numbers.

2. Experiment design

The design has three axes, each with its own controls.

- **Difficulty**: the benchmark ladder GSM8K, MATH-500, AIME 2024 (easy to hard), and synthetic problems where difficulty d is the exact number of dependent steps.
- **The written channel**: open (normal chain of thought or thinking), closed (the prompt demands only the final answer; we verify from token counts that the model really wrote nothing), capped (a hard token budget), or edited (we change one written value in the model's own work and let it continue).
- **The internal side**: J-space ablation (remove the currently active lens concepts from the activations, with a random-directions ablation of identical size as the control), a coarser residual-stream lesion, and a residual-stream patch (overwrite the internal state at one written token with a state of our choosing).

Interventions follow a fixed discipline. Ablation strength is chosen on neutral text before any task runs, by a rule stated in advance (the strongest dose keeping next-token agreement with the clean model at 0.80 or above and the perplexity ratio at 1.30 or below), so task effects cannot come from a dose that already breaks the model in general. Every cell logs three failure types separately, wrong answer, ran out of tokens, and no parseable answer, so accuracy effects are never confused with truncation. Predictions for each experiment were written down before it ran, with the opposite outcome stated as detectable.

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3. Experiment details

- **Assignment model and data**: Qwen3-4B, with the fitted public Jacobian lens for this exact model (neuronpedia/jacobian-lens). GSM8K test set; MATH-500; AIME 2024, meaning the 30 problems of AIME 2024 I and II as distributed in HuggingFaceH4/aime_2024. All revisions pinned in Reproducibility.
- **Synthetic tasks**, used where causal work needs exact ground truth for every intermediate value: variable chains (start with a 3-digit number, apply signed 2-digit additions and subtractions, report the final value; d = chain length), modular arithmetic chains, box tracking (5 boxes whose contents get swapped; state is parallel, not chained), and DAG path-following. Generators are seeded, and replay tests confirm the corruption arithmetic matches the generators.
- **Supporting models** for the mechanism experiments: Qwen2.5-7B-Instruct (also lens-fitted) and DeepSeek-R1-Distill-Qwen 1.5B/7B/14B white-box; DeepSeek V3.2, R1-671B, Llama 3.x up to 70B, and Claude Sonnet 4.5 through APIs, behavioral only.
- **Sampling**: temperature 0.6, top-p 0.95 for chain of thought and thinking; temperature 0 for direct answers. Fixed in advance.
- **Measurement calibration**: the externalization detector (does a known intermediate value appear in the trace) was checked by scoring each trace against a different problem's values; false positives are 0.00 to 0.08 on variable chains, but reach 0.72 on mod-97 chains at high difficulty, so mod-97 is excluded from externalization claims.
- **Replication before extension**: two known results were reproduced on our setup before building on it. Cutting a model's chain of thought short degrades accuracy the way the faithfulness literature reports (keep 60 percent of the trace and V3.2 scores 0.00; keep all of it, 0.97; Sonnet 4.5 goes 0.32 to 1.00), and a linear probe reads the final answer from the activations at R-squared 0.96 while a control probe stays at chance, validating the probe and patch machinery.

4. Experimental results and analysis

4.1. Accuracy with and without written reasoning

Models write nearly every intermediate value even on the easiest problems. On synthetic chains, the fraction of known intermediate values that appear in the written trace is 0.93 to 1.00 at every difficulty from 1 step to 64, for every model from 1.5B to 671B. Among correct traces at 16 or more steps it is exactly 1.000 (n=1935 across three model sizes), versus about 0.5 among wrong ones, and the ceiling is the same for early, middle, and late steps. If writing were just output formatting, models would sometimes skip early steps they could hold in their heads; they never do. We treat the 1.000 as a strong association rather than proof of necessity (the solution format asks for step results), and establish the causal role of the written values in the next subsection.

Forbidden from writing, models fail after one or two dependent steps. We prompt for the bare answer and verify from token counts that nothing else was generated:

The R1 distills kept generating reasoning despite the no-think template, so their cells are excluded: their channel was never actually closed. Internal capacity for chained computation is real but tiny, one to two steps for most models, about five for Llama-70B, and we cannot tell whether depth or parameter count sets it because the two grow together within model families. One scope note: these chains use random values that cannot be memorized. The same V3.2 that fails 2-step chains answers 38 percent of GSM8K correctly with one-word replies, because parts of natural problems can be recalled or recomputed in one step. That distinction, can the model recompute a value or not, turns out to govern everything below.

Qwen3-4B on the assignment benchmarks shows the same dependence on writing, and it grows with difficulty. Thinking-mode accuracy stays high while bare-answer accuracy sits near the floor on all three datasets. Two further patterns: told not to write, the model overruns its 32-token answer budget more often the harder the problem gets (0.29 of GSM8K attempts, 0.80 of AIME attempts), and the amount of thinking it writes when allowed grows steeply down the ladder (median 1952 tokens on GSM8K, 3503 on MATH-500, 11819 on AIME).

Writing is not overflow from a full workspace: the workspace never held a chain in the first place. But all of this is still consistent with two stories, a trace the model computes with, or a commentary alongside computation happening elsewhere. The editing experiments decide between them.

4.2. Editing the model's written values

Change one written value in the model's own work, and the final answer usually follows the change. We take a correct trace, replace one mid-chain value (say 417 with 457), and let the model continue. On the 7B reasoning model the answer follows the edit 31, 42, 50, and 36 percent of the time at d = 4, 8, 16, 32 (n = 149, 146, 138, 128 editable items of 150; the yield falls with difficulty because only correct traces are edited). If the trace were commentary, edits would do nothing. Two ambiguities remain, though: the edited text also feeds any fresh recomputation, and reasoning models re-solve the problem from scratch after their

Model	d=1	d=2	d=4	d=8	Really wrote nothing?
Qwen2.5-7B-it	1.00	0.96	0.04	0.00	Yes (median 4 tokens)
Llama-3.1-8B	1.00	0.47	0.00	0.00	Yes (2-4 tokens)
Llama-3.3-70B	1.00	1.00	0.93	0.03	Yes (2-4 tokens)
V3.2 (671B)	0.23	0.03	0.00	0.00	Yes (2 tokens)
R1 distills	-	-	-	-	No (74-89 percent overran)

Dataset	Bare answer acc (non-truncated)	Thinking acc (non-truncated)	Bare-answer overrun rate	Thinking cap rate
GSM8K (n=150)	0.10 (0.12)	0.84 (0.99)	0.29	0.46
MATH-500 (n=150)	0.11 (0.19)	0.65 (0.76)	0.47	0.23
AIME 2024 (n=30)	0.00 (0.00)	0.67 (0.90)	0.80	0.30

think block, which pulls answers back to the correct value without any internal memory being involved. That re-solve behavior is a finding on its own (the think phase reads the trace; the answer phase re-derives), and it is why the decisive version runs on a model that continues a worked solution directly.

Overwriting the internal state at the edited token controls the answer, whatever value we write into it. We overwrite the residual-stream activations at the edited token (middle layers) while leaving the visible text unchanged. On Qwen2.5-7B-Instruct at 10 steps (n=141):

The third-value row is the important one: the answer tracks whatever value sits in the internal state at that token, including values the model never wrote, computed forward through the remaining steps. So the state at a written token works like a memory slot that later computation reads. This also rules out the objection that restoring the correct state just injects the correct answer. On the reasoning model, the random-perturbation control is elevated (0.29, versus 0.00 on the instruct model) because of the re-solve behavior, but re-solving can only produce the correct answer, never the arbitrary third value, so the 0.74 there is clean. Restoring an early, middle, or late band of layers works about equally (0.96 / 0.97 / 0.94), so the value is stored redundantly across depth and we claim no specific circuit. Finally, a probe trained to decode the answer minus the known start value (the start inflates naive decodability) reads almost nothing two steps in (0.27) and 0.83 by the end: the answer comes into existence as the steps get written, not before.

Bigger models follow the edited value more often, not less. The same text-editing test through API prefill: R1-distill-7B 0.42, DeepSeek V3.2 0.78, Claude Sonnet 4.5 0.97. One might expect stronger models to hold more in their heads and lean on the trace less; the trend runs the other way. For anyone monitoring model reasoning by reading the trace, the trace stays load-bearing at the frontier. This is observational, since these models differ in far more than size.

Edits only matter when the model cannot recompute the value from the problem. On synthetic chains the follow-the-edit rate is high at every depth (0.80, 0.64, 0.78, 0.76, 0.70 at d = 3 to 16). On GSM8K it collapses to 0.10 against a 0.05 do-nothing floor (problems with 5 or more calculation steps, n=63): a GSM8K intermediate is usually one operation away from the given numbers, so the model recomputes it and ignores the edit. Written values act as memory for state the model cannot cheaply reconstruct, and not otherwise. We predicted the assignment-core edit test on Qwen3-4B would therefore also sit near its floor. That prediction failed, and the failure sharpened the claim: on Qwen3-4B the answer follows the edit on 0.87 of GSM8K worked solutions (n=84, resample floor 0.00, no unparseable answers). Putting the two GSM8K results side by side, the 671B model recomputes these intermediates and ignores edits (0.10) while the 4B model reads them back (0.87), so recomputability is relative to the model's internal capacity, not a property of the task alone: a value is read back when recomputing it exceeds what the model can do in its head, and what a frontier model recomputes, a small model must reread. This also fits the capability trend above. On chain values no model can recompute, stronger models follow the trace more faithfully; on values they can recompute, they stop depending on the trace.

4.3. Token budgets and scratchpad formats

Under a token budget, models shorten their wording but keep writing every value. With hard output caps (V3.2, chains at d=16), traces compress up to 2.5x with accuracy intact, and then fail completely once the cap cannot fit the values themselves, rather than shifting any computation inward:

The failure floor sits near 12 tokens per step; below it, the fraction of values written simply tracks budget over need (0.72, 0.36, 0.17 as difficulty rises at budget 128). All wall failures are genuine truncations, not grading artifacts (0 of 60 lucky matches). One caveat: the model is told its cap, so failing to compress and failing to plan are confounded.

Only the computed values matter; writing operations

Intervention	Outcome
Edit the text only	Answer follows the edit on 0.84 of items
Also restore internal state to the correct value	Answer returns to correct: 0.97 [0.94, 0.99]; at d=20: 0.93
Instead write a third, never-written value into the state	Answer follows that third value: 0.76 [0.70, 0.82]
Write a random perturbation of the same size	Answer returns to correct: 0.00 [0.00, 0.01]
Same experiment on the reasoning model (n=91)	Correct-restore 1.00; third value 0.74 [0.66, 0.82]

Budget (d=16)	Tokens used	Accuracy	Values written
Unlimited	466	1.00	1.00
256	184	0.93	0.98
128	128 (hit cap)	0.00	0.72

without their results fails. Asking for the same chains in different formats: code-style lines with each computed value succeed at d=48 in 555 tokens; ordinary prose succeeds in 1105; a verbose format restating all values every step collapses (0.07 accuracy at d=48); and a code format instructed to omit computed values succeeds only where the model disobeys and writes them anyway, with per-item value-writing correlating with correctness at +0.74 (0.58 accuracy when under half the values get written, 1.00 when nearly all do). Efficient scratchpads and legible scratchpads coincide: both are the compact value-carrying kind.

4.4. Serial versus parallel tasks, and the J-space ablation

A task whose state is many independent values fits in the model's head; a task whose state is one long dependency chain does not. Box tracking (5 boxes, repeatedly swapped contents) needs as much state as a chain but with no step-to-step dependency, and it behaves opposite to chains: the model tracks 16 swaps correctly while writing only about a fifth of the state, and writing does not predict correctness there (0.20 among correct versus 0.17 among wrong, where chains sit at 1.00 versus 0.5). The causal check agrees: a residual-stream lesion at matched dose cuts box-tracking accuracy by 0.34 (target window) and 0.33 (control window), against 0.11 and 0.04 for chains, at difficulties where both tasks have room to fall. The internally-stored task is about three times more fragile to internal damage, whichever window is hit. This lesion is blunt, though: it also damages the model's re-reading of written values, visible at longer chains (a 0.38 drop at d=8), which is exactly why the targeted J-space version matters.

Ablating the active J-space concepts, at a dose verified harmless on neutral text, changes nothing on these tasks. The ablation removes, at every token position, the top-16 concepts the lens reads as currently active (strength 0.5, the maximum passing the pre-frozen harmlessness rule; next-token agreement 0.87 and perplexity ratio 1.12 on neutral text, with the random-directions control arm at the same size actually more disruptive on neutral text, ratio 1.29, which works against us finding a spurious targeted effect). Under this ablation, bare-answer and chain-of-thought accuracy both match the unablated model at every difficulty, and the amount the model writes does not change. So either the decision of what to write is fixed by training rather than responsive to internal load, or chain state lives outside the top lens concepts. The assignment-core 2x2 (ablation by answer-mode on GSM8K, Qwen3-4B, same frozen-dose discipline) tests which, on the benchmark where the original J-space ablation asymmetry was reported [result pending]. A dose escalation with neutral-text damage reported alongside is the natural follow-up.

5. Positioning

Expressivity theory says fixed-depth transformers can only do limited serial computation in one forward pass and that chain-of-thought length buys serial power (Merrill and Sabharwal 2023, 2024; Li et al. 2024; Feng et al. 2023); it predicts our behavioral results and motivates the mechanism experiments, which it does not itself describe. Faithfulness studies (Turpin 2023; Lanham 2023; Bentham 2024) and recent work decoding reasoning state from activations (2603.05488, 2603.01437, 2604.18307, 2606.13603) establish the starting point that traces and internal state can disagree. The two closest papers stop short of the causal test run here: 2604.15726 calls for a token-corruption experiment it never runs, and 2605.30343 builds latent memory blocks without manipulating capacity or measuring readback. The J-space construct and the lens are from Gurnee et al. 2026.

6. Negative Results

No writing threshold exists, which killed the original hypothesis. The project was designed around finding the difficulty at which models start externalizing, and a whole phase was budgeted for fitting a scaling law to that onset across model sizes. There is no onset to fit: writing is saturated from one-step problems at every scale, so the law had nothing to describe. The salvageable quantity is on the other side of the boundary, the depth a model can handle with no writing at all, and there the data supports only an ordinal statement (one to two steps for most models, about five for Llama-70B) because depth and parameter count grow together within model families and cannot be separated with public checkpoints. This null is what turned the project from measuring a threshold into finding the mechanism: if writing is always on, the interesting question becomes what the writing is for.

Making the internal side scarcer does not make the model write more. If externalization were a response to internal pressure, induced scarcity should induce writing. We tested this twice with increasingly fair instruments. A heavy residual lesion pushes the model past an accuracy cliff, and there writing degrades along with everything else, traces disintegrate rather than expand. The objection that the lesion was simply too crude motivated the calibrated J-space ablation, a targeted squeeze verified harmless on neutral text, and it produces no change either: not in accuracy, not in the amount written, not in the fraction of values externalized. Two readings survive: the write policy is set by training and does not respond to inference-time load, or chain state is carried outside the top lens concepts. Either way, the adaptive-hierarchy picture, in which models manage a live trade-off between workspace and page, has no support in these experiments; the allocation appears static in both directions, since capping the page also fails to push computation inward (the budget wall is truncation, not internalization).

The GSM8K edit test came back null, and the explanation became the project's scoping result. Corrupting a written intermediate on GSM8K changes the answer at roughly the sampling noise floor (0.10 versus 0.05), where the same edit on synthetic chains flips the answer most of the time at every depth. The difference is not depth, the chain flip rate is flat from three to sixteen steps, it is that a GSM8K intermediate sits one operation away from numbers still visible in the problem, so the model recomputes it and the edit is irrelevant. This null defines the mechanism's boundary: written values function as memory precisely for state the model cannot cheaply reconstruct, and a large fraction of benchmark math does not have that character. Any claim about chain-of-thought as working memory that does not carry this qualifier is overstated, including ours before this test.

Two early claims were withdrawn after a confound was found. A clean-return rate of 0.48 to 0.68 after edits, and its rise at high difficulty, were initially read as the model verifying written values against a persistent internal copy. Both readings collapsed on the observation that the full problem statement stays in context, so returning to the correct answer requires no internal copy at all, only re-derivation from the prompt; the reasoning-model experiments then showed exactly that behavior (the post-think phase re-solves from scratch). An early-decodability claim fell the same way: the answer seemed readable from activations two steps into a chain, but only because the chain's start value dominates the answer's variance, and regressing it out shows the computed part is absent early and builds as steps are written. No internal-copy or verification claim survives, and the corrected probe is evidence for the tape picture rather than against it.

Smaller instrument failures, recorded for completeness: the DAG task cannot measure externalization because every node name already appears in the prompt; mod-97 chains were dropped when the permutation control showed match false positives up to 0.72; and the first lesion experiment was invalid as coded (it fired only during generation, leaving the bare-answer condition nearly unlesioned) and was rerun after the fix.

Limitations. The central mechanism applies to state the model cannot recompute, which excludes much of GSM8K-style math. Synthetic tasks trade realism for exact ground truth. White-box causal results are at 4B to 7B scale; frontier evidence is behavioral. The patch overwrites the full residual at a position, and although the third-value result shows the value is what matters, the value's own subspace is not isolated; splitting the patch into its J-space component and the remainder is the planned experiment that would locate the memory slot relative to the lens-readable workspace. The lesion cannot separate workspace damage from read-back damage. Remaining open: a within-task recomputability manipulation, that patch decomposition, and the pending Qwen3-4B cells, which either reproduce the original ablation asymmetry on its intended benchmark or extend our null to it.

7. Reproducibility

Pinned revisions:

- Qwen/Qwen3-4B 1cfa9a720891; Qwen/Qwen2.5-7B-Instruct a09a35458c70
- DeepSeek-R1-Distill-Qwen-1.5B ad9f0ae0864d; 7B 916b56a44061; 14B 1df8507178af
- neuronpedia/jacobian-lens a4114d7752d1 (qwen3-4b and qwen2.5-7b-it artifacts); lens library commit 581d398613e5
- GSM8K test.jsonl sha1 4a3eef48d603; HuggingFaceH4/MATH-500 6e4ed1a2a79a; HuggingFaceH4/aime_2024 2fe88a2f1091

Dose rules were frozen before task cells ran. Every cell logs cap hits, unparseable answers, and bare-answer token counts. Raw data regenerate from src/harness; figures from src/analysis/figures.py; the dated evidence trail is notes/results-log.md.